

UAV-X: Resilient BVLOS Swarm Challenge

Stage 1 – Preliminary Design Verification: Technical Proposal

Team ID	TM-B02FB9B1E5C
Team Name	Useless Engineers
Team ID / Registration No.	UAVX-4B327D91E51E
Institution / Organization	Indian Institute of Technology Madras
Team Leader	Swapneel Banerjee · technoswapneel@gmail.com · 8420317195
Members (up to 5)	Swapneel Banerjee, Soham Bhattacharyya
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1. Abstract / Executive Summary

An earthquake or landslide has cut terrestrial communications. A fleet of UAVs flying at 5 m/s with a 100 m radio range must find and report 10 Points of Interest (Pol)s in a 1 km × 1 km area. Each Pol must be reported within 10 s of discovery, and the UAVs must stay linked to a Ground Control Station (GCS) outside the area through a multi-hop aerial network. Our design treats connectivity as a hard constraint on motion rather than a quantity to be monitored. The primary strategy is a **T-sweep**. A line of 12 UAVs spaced 80 m apart sweeps the area back and forth, and an 11-relay trunk dropped along the middle row links it to a sentinel UAV at the edge of the area, the only node within range of the GCS. A **relief scheduler** plans each UAV's departure backwards from its 20-minute battery deadline through a single-lane exit. An alternative **adaptive strategy** adds a shared coverage map, a deadline-derived tether ($R + v(T - t_{\text{comms}} - t_{\text{confirm}}) \approx 140\text{--}150\text{ m}$) and make-before-break relay rewiring. In **500 Monte Carlo trials** of the T-sweep, 148 of them with a random UAV failure injected, 95.8% of trials reported all 10 Pol)s. Every detected Pol was reported within 10 s, the packet delivery ratio was 100%, connectivity availability averaged 95.5% (99.5% without failures) and there were no collisions. The swarm recovered from 77.7% of injected failures, with a median recovery time of 112 s. Failed trunk relays that were not replaced are the main remaining weakness. Fixed-fleet sizing of the adaptive strategy shows that 58 UAVs is the smallest fleet that succeeds on all five layouts tested so far. The novel elements are the deadline-derived tether, battery-deadline relief scheduling and fleet sizing under a hard fleet cap.

2. Problem Understanding & Mission Analysis

2.1 Mission scenario and assumptions

A GCS (called OPS in our simulators) is set up outside the affected area. Every UAV takes off from it and enters and leaves the area through fixed lanes. Surveying UAVs sweep the area for Pol)s and report each one the moment it is detected. Reports travel hop by hop through relay UAVs to a sentinel UAV posted at the edge of the area, which is the only link to the GCS. Our simulators use the following assumptions.

Item	Assumption used in our simulators
Map	1000 m × 1000 m open field, modelled as 25 × 25 cells of 40 m. The Python port uses continuous positions and keeps the grid only for coverage and route planning.
GCS / entry	T-sweep: OPS 75 m left of the area, level with its middle. The sentinel holds the middle cell of the left edge, 95 m from OPS. There is an arrival lane (cell above the sentinel) and a departure lane (cell below). Adaptive modes: sentinel posted just inside a 3-cell gate at the top-middle.
UAVs	T-sweep: 24 posts (1 sentinel, 12 line, 11 trunk) plus relief UAVs. Fleet-sizing studies cover caps of 33–90 UAVs.
Speed / tick	5 m/s. Grid simulators: one cell move per 8 s tick. Python port: 1 s tick (5 m per tick).
Endurance	20 min maximum flight. A UAV does not fly again after landing. The mission is bounded so that every UAV has landed by 45:00.
Comms range	100 m per hop, multi-hop. A link exists if two nodes are within 100 m.

Item	Assumption used in our simulators
Polis	10 Polis at random cells, more than 2 cells apart and away from the entrance cells. They appear at random times within the first 10 min and stay once they appear.
Sensing	A UAV detects a Pol within 1 cell of its position (3×3 -cell footprint). Only UAVs linked to OPS act as sensors.
Separation	20 m minimum, checked continuously (Python port). Grid simulators allow at most one UAV per 40 m cell.
Report deadline	10 s from first detection to receipt at the sentinel/GCS.

2.2 Requirements and objectives

The table maps each requirement to how the design meets it and how our simulators check it.

Requirement	How the design addresses it	How it is checked in simulation
Survey all assigned Polis	The sweep line covers the full width back and forth. Adaptive mode visits the nearest unseen cell, then the stalest cell.	Recall when all 10 Polis are reported (or at the latest safe time). Polis reported are counted per trial.
End-to-end connectivity with the GCS	Trunk / MST relays at ≤ 100 m spacing. The line only advances if every line UAV stays linked.	BFS reach-set from OPS every tick. Only linked UAVs sense.
Dynamically assign relays	Trunk drop-off behind the line. Hull $\rightarrow$ MST relay slots are recomputed with reuse-first assignment.	Relay slots and roles are logged per tick.
Reconfigure on degradation, failure or RTH	Relief hand-over at the post, make-before-break merges, connectivity-preserving moves.	One random UAV failure (line, trunk or sentinel) injected in 148 of 500 trials. Recovery time and efficiency are logged.
Prioritise new high-priority regions	Polis appearing mid-mission are handled (§4.2). No priority re-planning yet.	A priority-weighted completion score is logged for every trial.
Safe, on time, within geofence and charge	Separation rule, grid-bounded motion, battery-deadline relief planner, recall by 45:00.	Rule-break counters: flights over 20 min, landings after 45:00, late reports.

2.3 Key challenges and trade-offs

- **Coverage vs connectivity.** The far corner is $\sqrt{(500^2 + 1000^2)} \approx 1118$ m from a top-middle entrance, which is about 224 s of flight at 5 m/s. Multi-hop transit over about 12 hops takes well under 1–2 s, so the limit is flight time, not latency. A discoverer must therefore already be connected, or nearly so, when it detects a Pol. We address this with the tether rule (§5.3) and by keeping the whole sweep line linked.
- **Report-only vs persistent links.** A persistent link to every Pol is the largest drain on the fleet. A finished MST network needs 27.8 UAVs on average (33 at the 95th percentile). Building straight chains before merging needs 57.8 on average (71 at the 95th percentile). The T-sweep therefore reports and moves on.
- **Endurance vs relay allocation.** Every post must be relieved before its UAV's battery deadline, and entry and exit share single lanes. The departure schedule is planned backwards from the deadlines (§4.3).
- **Fleet size vs churn.** A larger fleet is not always faster. On one layout, 53 UAVs finished in 33 min but 63 or more took 135 min, because spare UAVs were spent on relay churn.
- **Discretisation.** With 40 m cells, legal hops shrink to 80–89 m, and an 8 s tick is only 1.25 ticks of the 10 s deadline. This motivated the continuous-space, 1 s-tick Python port.

3. System & Software Architecture

3.1 System overview

The system has four physical tiers linked by multi-hop radio: the GCS/OPS, the sentinel UAV, the relay layer and the survey UAVs with their sensing payload. An autonomy stack plans goals and roles each tick from the UAVs' positions, detections and battery states (Figure 1).

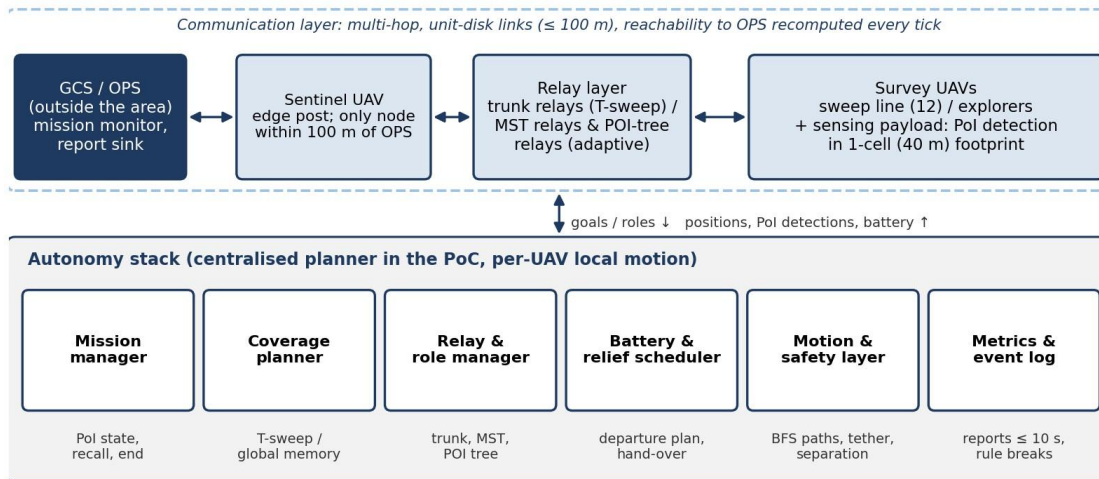


Figure 1: System architecture.

3.2 Software architecture

The proof-of-concept (PoC) is written in **Python** for the simulation core and experiments, with **HTML/JavaScript** canvas pages for interactive runs and replay. No robotics middleware is used in the PoC; all modules run in-process. The key libraries are NumPy, SciPy, NetworkX and Shapely (core), pandas, Matplotlib and tqdm (experiments), and PyYAML and pytest (config and tests). Each tick, the planners read the shared world state and produce a goal per UAV. The motion and safety layer turns these goals into collision-free, connectivity-preserving steps, and the new poses are written back to the world state (Figure 2). Headless runners execute seeded batches and write summary metrics and JSON traces.

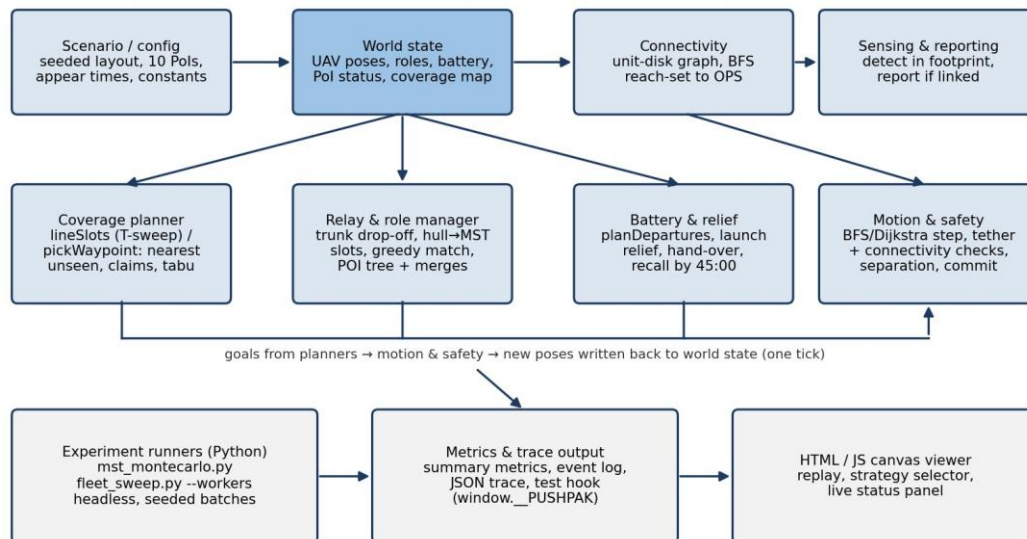


Figure 2: Software module and data-flow diagram.

3.3 Autonomy pipeline

Each tick, every UAV runs the same loop. **Perception:** detect any appeared POI within its footprint, but only if the UAV is linked to OPS. **Situational awareness:** update the shared coverage map (seen or staleness per cell), the connectivity graph and battery states. **Task/role assignment:** T-sweep posts are sentinel, line, trunk or relief. Adaptive roles are explorer, relay or anchor. **Planning:** compute a role-specific goal and a BFS/Dijkstra path around other UAVs, checked against the tether and connectivity rules. **Control:** a one-step move, with conflicting moves resolved in a fixed order so that no two UAVs occupy the same place. In the PoC, coordination is **centralised** at OPS over shared state. This keeps one global coverage map, a single relay plan and a single departure schedule. Because connectivity is enforced, every UAV's state is available to the planner.

4. Swarm Autonomy & Mission Planning

4.1 Task allocation and coverage

T-sweep (primary). Twelve line UAVs hold rows 1, 3, ..., 23 (80 m apart) and advance one column at a time across the area and back. With a 3-cell-wide footprint, each pass covers the full height. The line advances only when every line UAV is in its slot and the next step keeps every line UAV linked to OPS. If another UAV occupies a slot, the line UAV takes the nearest free cell in the same column. The line forms centre-outward, so each new UAV flies along UAVs already in place. **Adaptive global-memory search (alternative).** The swarm shares one map of every cell ever seen. Each explorer heads for the nearest unseen cell inside a 140 m frontier around the connected network. If none is left, it picks the unseen cell with the least combined effort (network growth plus flight). Once everything has been seen, it re-checks the stalest ground. Explorers claim waypoints at least 2 cells apart, skip a waypoint that cut them off for 30 ticks, and retarget instantly if another UAV covers their cell first. A structured sweep and a boolean coverage map replaced the original Brownian search, which suffered from a coupon-collector effect on the last cells.

4.2 Dynamic re-tasking and priority handling

Pols appear at random times during the mission, and a cell that has already been seen can gain a Pol later. The sweep line therefore keeps going back and forth, and the adaptive search keeps re-checking the stalest ground, until all 10 Pols are reported. In the hold variant (T-sweep + hold), the line UAV that finds a Pol stays on it. One of two reserve UAVs trailing the line takes its slot, and a new reserve is sent in. The T-sweep does not yet re-plan for higher-priority Pols, and its outcome is measured with a priority-weighted completion score. That score averaged 97.4% over 500 trials, 100% in every trial without a failure and 91.1% in trials with a failure (§10).

4.3 Endurance and recharging management

Endurance is tracked per UAV from take-off (20 min, or 150 ticks of 8 s). Each post's UAV is relieved by a fresh UAV launched early enough to take over before the incumbent must fly home. The departure lane releases one UAV per tick, so the scheduler gives every UAV that needs relief its own lane slot. Slots are planned backwards from each battery deadline, with the latest deadline getting the latest slot. UAVs whose deadlines clash leave slightly earlier instead of queueing together. The travel-home estimate is $1.8\times$ the straight-line cell count plus 4 ticks (the measured worst case was about $1.7\times$), with 3 ticks of spare battery on landing. The relief is launched to arrive just before the scheduled departure, and the launch queue is ordered by urgency. The hand-over happens when the relief is within one cell of the post. A safety net forces a UAV home if its remaining flight time is within 4 ticks of its return time. Reliefs in transit and UAVs flying home do not search. The swarm is recalled once all Pols are reported, or at the latest time that still lets every UAV land by 45:00. UAVs do not fly again after landing, so recharging and reuse are not modelled.

5. Resilient BVLOS Communication

5.1 Communication model and assumptions

Link model: unit disk. Two nodes are linked if their Euclidean distance is ≤ 100 m. A UAV is *connected* if a multi-hop path joins it to OPS, which is computed by BFS every tick. **Latency:** per-hop delay is assumed so that about a dozen hops take roughly 0.1–1 s in total, and this budget is carried into the tether rule (§5.3). Each Pol report is sent as one packet along the multi-hop path to OPS, so packets sent equal Pols found. Link bandwidth is not modelled in the PoC. The organizer-provided standardised communication assumptions will be adopted once they are released.

5.2 Multi-hop relay network design

T-sweep topology. OPS → sentinel → trunk (row 12, columns 2, 4, ..., 22, spaced 80 m) → sweep line (itself a chain of 80 m hops). The trunk relays launch once the line starts moving. They trail in single file along the middle row behind the line, and each one drops at its column once the line has passed. Relays therefore sit where the search already went, with no transit lag. **Adaptive topology.** Every 5 ticks the planner takes the convex hull of the searchers and builds an MST from the sentinel to the hull UAVs. It places relays only where an MST edge exceeds 100 m, at 2.35-cell spacing, just under the link limit. Existing relays are reassigned first. Fresh UAVs are recruited only for a genuine shortfall, and surplus relays become explorers again. In the hold variant, each Pol holder links to the nearest node of the permanent network, so the network grows like an MST. **Measurement.** The reach-set and the number of connected components are computed every tick. Over 500 T-sweep trials, the packet delivery ratio was 100% for every report sent. The logged packet latency averaged 9.95 latency ticks, with a maximum of 26. Connectivity availability averaged 95.5% (median 99.1%).

5.3 Connectivity-aware planning

The report deadline gives an explicit **tether**: $\text{tether} = R + v \times (T - t_{\text{comms}} - t_{\text{confirm}})$. An explorer may be at most this far from a stable connected UAV (the sentinel, a settled relay or an anchor), because it can then fly back into range and report in time. With $R = 100$ m, $v = 5$ m/s and $T = 10$ s:

Assumption	Tether
Instant comms and detection	150 m
1 s comms budget	145 m
1 s comms + 2 s confirmation	135 m

The practical design value is 140–145 m, and the adaptive frontier is set at 140 m. The rules enforced in the Python port are as follows. (i) An explorer's move is legal only if it ends within 150 m of a stable connected UAV, and an explorer already outside that range may only move closer. (ii) No move may disconnect a stable connected UAV or push a tethered explorer outside its tether. (iii) A reassigned relay still counts as infrastructure on its old spot until nothing depends on it. (iv) Merges are make-before-break: the anchor switches to a new branch only once the branch is manned, and the old chain is then pruned. In the T-sweep, a linked line UAV never steps out of range while the line forms.

6. Autonomous Relay & Role Management

T-sweep roles are fixed posts: sentinel, line and trunk. Role changes happen only at relief. A relief UAV flies to the UAV it replaces and takes over the post, and the incumbent switches to the returning role and leaves by the departure lane. Once dropped, trunk relays stay in place, so relay reallocations occur only at hand-overs. **Adaptive roles** are interchangeable. A relay that spots a Pol breaks off to claim it, and surplus relays are redeployed as explorers or sent home once everything is found. A UAV that cannot join a chain holds its position for up to 2 minutes and moves into any slot that opens, otherwise it returns. Under a hard fleet cap, recruitment follows a fixed priority: POI-tree relays, then exploration relays, then explorer top-up, with freed and holding relays reused first. An unmanned POI-tree slot takes the nearest searching explorer, but at least one explorer keeps searching while Poles remain. Role decisions are made centrally (§3.3). **Measured relay reallocations** (500 T-sweep trials): none in 475 trials, including all 352 without a failure. The remaining 25 trials, all of which had an injected failure, needed 3–20 reallocations (mean 0.52 per trial overall).

7. Safety & Collision Avoidance

Maps to Safety & Collision Avoidance – 10% of the score.

Separation. In the Python port, the 20 m minimum is checked continuously along each move's whole path against every other UAV's path. In the grid simulators, every UAV plans a BFS path that treats occupied cells as blocked. Moves are committed in a fixed order, and any move into a cell already taken is rejected. Poles are kept away from the gate so that a UAV holding a Pol can never block entry. **Geofence.** All motion is confined to the area's cell bounds, and UAVs enter and leave only through the designated lanes. **Charge safety.** The relief planner and its safety net (§4.3) bring every UAV home with spare battery. The simulator counts any flight over 20 min and any landing after 45:00 as a rule break. **Measured:** the minimum separation was 20.001 m in the Python port. In 500 T-sweep trials it was 40 m (one UAV per cell) with 0 collisions. No trial without a failure broke a charge or landing rule. 25 trials with a failure had flights over 20 min, and 12 of them also landed after 45:00 (§10).

8. Innovation & Technical Merit

Maps to Innovation & Technical Merit – 5% of the score.

- **Deadline-derived tether.** The allowed distance between an explorer and the network comes directly from the 10 s report deadline, $R + v(T - t_{\text{comms}} - t_{\text{confirm}})$. This turns a timing requirement into a geometric constraint enforced on every move.
- **Connectivity as a hard rule.** Connectivity-preserving moves and make-before-break rewiring replace the usual approach of monitoring disconnection and recalling UAVs afterwards.
- **Battery-deadline relief scheduling.** Lane slots are planned backwards from each UAV's battery deadline, so relief UAVs arrive just in time through single-lane entry and exit.
- **Drop-off relay deployment.** Trunk relays trail the sweep line and stop where the search has already been, so relays are placed with no transit lag.

- **Evidence-based fleet sizing.** A static MST Monte Carlo (1,000 layouts) sets a lower reference. A fixed-fleet sweep with a hard cap measures the smallest fleet that actually succeeds.
- The design builds on Andrew's monotone-chain convex hull [1], minimum spanning trees [2] and shortest-path search [3].

9. Simulation Framework & Proof-of-Concept

Required deliverable — a working proof-of-concept simulation.

9.1 Simulation setup

Frameworks: custom open-source simulators, with a Python core (`sector_sweep.py`, `fleet_sweep.py`, `mst_montecarlo.py`) and HTML/JS canvas simulators (`PUSHPAK - T sweep, report only` and `Sector Sweep (global memory)`). The Sector Sweep page provides four strategies: adaptive search with Pols held, adaptive search (report only), a formation sweep (report only) and T-sweep + hold (MST network). **Scenario:** as in §2.1.

Communication model: as in §5.1. Each trial writes one row of 36 fields covering mission, communication, autonomy, robustness and safety metrics (`pushpak_runs_raw.csv`). Aggregate statistics (n, min, mean, median, mode, 5th and 95th percentile, max) go to `pushpak_summary_stats.csv`. These fields will be mapped to the organizers' benchmark scenarios and standardised log format when they are released.

9.2 Proof-of-concept results

Static network sizing (MST over sentinel + 10 Pols, $\lceil L/100 \rceil - 1$ relays per edge, drones = relays + 10 anchors + 1 sentinel, 200 trials per size; 1,000 trials at 25×25):

Grid (arena)	Mean	Std	Median	95th %	Max	Mean MST
10 × 10 (400 m)	14.4	1.4	14	17	19	0.90 km
15 × 15 (600 m)	18.8	1.8	19	22	23	1.32 km
20 × 20 (800 m)	23.0	2.4	23	27	30	1.74 km
25 × 25 (1000 m), 1,000 trials	27.8	2.9	28	33 (CI 32–33)	36	2.20 km

Fixed-fleet sweep (success = mission complete within 240 min, every report within 10 s and no separation violations; battery ignored; same 5 seeded layouts at every cap). The uncapped peak demand was 61–67 UAVs.

Fleet cap B	33	38	43	48	53	58	63-90
Successes (of 5)	0	1	2	4	4	5	5
Failures	5 timeout	3 timeout, 1 late	2 timeout, 1 late	1 late	1 late	—	— (cap no longer binds)

T-sweep Monte Carlo (scenario S3, 500 trials, random layouts and Pol appearance times). In 148 trials one UAV failed at a random time (observed ticks 35–111): 71 line, 71 trunk and 6 sentinel failures.

Metric	No failure (352)	One failure (148)	All (500)
Trials with all 10 Pols reported on time	352 (100%)	127 (85.8%)	479 (95.8%)
Mission duration, mean / median / p95 (min)	16.3 / 16.0 / 18.3	23.6 / 18.9 / 58.5	18.5 / 17.3 / 21.9
UAVs launched, mean / median / p95	33.8 / 36 / 36	39.2 / 37 / 70.7	35.4 / 36 / 40
Connectivity availability, mean (%)	99.5	85.8	95.5
Communication downtime, median / p95 (s)	4 / 8	96 / 871	8 / 160
Trials with any rule break	0	25	25
Priority-weighted completion, mean (%)	100.0	91.1	97.4

Failure recovery. The swarm recovered in 115 of 148 failure trials (77.7%): 69/71 line, 6/6 sentinel and 40/71 trunk failures. The median recovery time was 14 ticks (112 s), and the median reconfiguration efficiency was 85.7%. Most failed recoveries were trunk failures whose relay slot was never re-manned. In these trials the fleet grew to 70–83 UAVs and missions ran past 45:00. A few line and sentinel failures were flagged as recovered, yet connectivity fell to about 16–19% and only 0–3 Pols were found. Both cases are the priority for Stage 2.

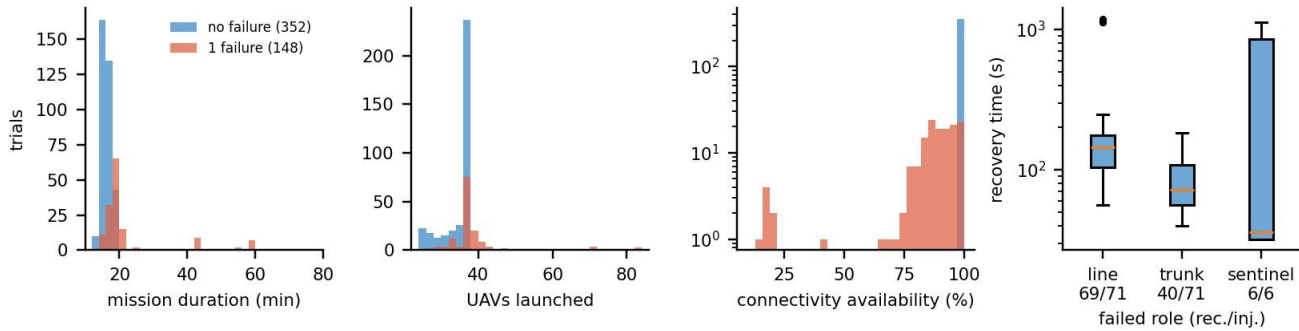


Figure 3: T-sweep results over 500 trials. Distributions of mission duration, UAVs launched and connectivity availability (blue: no failure; red: one injected failure), and recovery time by failed role.

10. Evaluation & Performance Metrics

S3 (primary): T-sweep simulator, 500 Monte Carlo trials, 148 of them with one injected UAV failure (§9.2). Values are means over all 500 trials unless stated. **S1**: continuous-space Python port of the adaptive strategy (version 1, staleness search). **S2**: fixed-fleet sweep (§9.2).

Category	Performance metric	Your result
Mission	Completion rate	95.8% of trials report all 10 Pols; 97.4% of Pols reported — S3 (100% without failure)
Mission	Completion time	18.5 min mean (median 17.3, p95 21.9) — S3
Mission	Priority-weighted mission score	97.4% mean (100% without failure, 91.1% with) — S3
Communication	Packet delivery ratio	100% — S3
Communication	Latency	9.95 latency ticks mean, 26 max (simulator comms-model units) — S3
Communication	Connectivity availability	95.5% mean, 99.1% median (99.5% without failure) — S3
Communication	Communication downtime	64 s mean, 8 s median, 160 s p95 — S3
Autonomy	Relay reallocations	0.52 per trial (0 in 475 of 500 trials) — S3
Autonomy	Recovery time	112 s median, 177 s mean (115 recovered failures) — S3
Autonomy	Network reconfiguration efficiency	85.7% median, 77.9% mean — S3
Robustness	Performance after UAV / link failures	77.7% of failures recovered; 85.8% of failure trials report all 10 Pols; availability 85.8% — S3
Safety	Collision count	0 — S3 and S1
Safety	Minimum inter-UAV separation	40 m (one UAV per cell) — S3; 20.001 m — S1

Charge: no UAV flew beyond 20 min in any of the 352 trials without a failure (longest flight 15.3 min). In 25 of the 148 failure trials, 1–4 UAVs exceeded 20 min, and 12 of these trials also had landings after 45:00. **Geofence:** motion is confined to the area's cells by construction, and UAVs enter and leave only through the lanes. The logger has no separate geofence counter.

11. Reproducibility

Required deliverables — source code, installation instructions, demonstration video.

11.1 Source code]

File	Contents
requirements.txt	numpy, scipy, networkx, shapely, pandas, matplotlib, tqdm, pyyaml, pytest (optional: numba, seaborn)
sector_sweep.py	Continuous-space simulator with tether, connectivity-preserving moves and make-before-break

File	Contents
mst_montecarlo.py	Static MST drone-count Monte Carlo
fleet_sweep.py	Fixed-fleet success-rate sweep (imports sector_sweep.py; --workers for parallel runs)
PUSHPAK – T sweep, report only (.html)	T-sweep simulator with relief scheduling; headless test hook window. _PUSHPAK
Sector Sweep, global memory (.html)	Multi-strategy simulator (adaptive held / report-only, formation sweep, T-sweep + hold)
pushpak_runs_raw.csv / pushpak_summary_stats.csv	Per-trial metrics (500 T-sweep trials, 36 fields) and summary statistics reported in §9.2 and §10

11.2 Installation & run instructions

- *Step 1 – environment setup:* OS—Linux/Windows; Python version 3.13
- *Step 2 – build / install:* `pip install -r requirements.txt`. The HTML simulators need no installation.
- *Step 3 – run the demo scenario:* open either HTML file in a web browser and press Start. Batch sweep: `python fleet_sweep.py --workers`

11.3 Demonstration video

Link: https://drive.google.com/drive/folders/1D6tUhoQQd3vU_SswoGfocFpWMTrXx3PK?usp=sharing

12. Team & Contributions

Member	Role / contribution
Swapneel Banerjee (Team Leader)	Multi Agent Optimization
Soham Bhattacharyya	Multi Agent Architecture

13. References

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